Muhammad Talha Zafar

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ABOUT ME

I am a first-year **Mechanical Engineering** student at **NUST** Islamabad with a strong academic foundation and a passion for precision design. Having recently completed my foundational semesters with a focus on problem-solving and technical drawing, I am eager to transition from theory to practice at NASTP. I am driven by a deep interest in aerospace innovation and am seeking an internship where I can contribute to technological growth while immersing
myself in real-world engineering projects alongside industry leaders.

EDUCATION AND TRAINING

Matriculation [2020 – 2022]

Sanabil Educational Complex, Dad Fatyana, Chichawatni

Field(s) of study: Science | **Final grade:** A+

Intermediate [2023 – 2025]

Govt. Comprehensive Higher Secondary School, Sahiwal

Field(s) of study: Pre-Engineering | **Final grade:** A+

Bachelor of Mechanical Engineering

National University of Science and Technology, Islamabad

Field(s) of study: Mechanical Engineering

PROJECTS

Automatic Parking System

[2025 – Current]

Designed and prototyped an automated parking solution using Arduino Uno and IR sensors to detect vehicle occupancy. Integrated Servo motors for automated gate control and an I2C LCD for real- time slot availability updates.

IoT-Based Automated Irrigation System

Engineered a self-regulating watering system centered on an Arduino Uno and Soil Moisture Sensors (Hygrometers). Programmed a system in C++ to trigger a 5V Relay and DC Water Pump based on real-time soil data.

Rube Goldberg Engineering Challenge

Developed a Rube Goldberg machine utilizing principles of energy transfer and mechanical advantage. Designed a sequence of complex chain reactions to complete a simple task, demonstrating proficiency in physics, structural stability, and iterative prototyping.

Driver Drowsiness Alarm

Prototyped wearable glasses with IR sensors to detect eye closure and microsleep. Designed a C++ logic loop to trigger an audible alarm, enhancing road safety via real-time monitoring.

Autonomous Line-Following Robot (LFR)

Engineered a high-speed autonomous robot using an Arduino Uno and an IR Sensor Array for precision path tracking.

SKILLS

Microsoft Office | Microsoft Word | Microsoft Excel | Microsoft Powerpoint | Website Development | Programming | Arduino | Machine Learning | Good Communication | Engineering Drawing

Programming Languages: C | C++ | HTML | CSS | Java | Python
Software: AutoCAD | Arduino IDE | ROS 2 | VS Code | Platformio | Proteus |Dev C++

HONOURS AND AWARDS

STEM Competition Winner for the District Sahiwal [2025]